

Hybrid integration of bare die active electronics, wireless power and communication elements on ultrathin implantable substrates

Pietro Palopoli¹, Mingxiang Gao², Marta S.D.S Vasconcelos¹, Simon Seyfert¹, Julian V.J.R.Rodriguez¹,Rodrigo A.S.D.Anjos¹, Ivan Furfaro¹, Giusy Spacone³, Sebastian Frey³, Horacio Londono Ramirez¹, Victor Druet¹,Andrea Cossetti³, Luca Benini³, Giuseppe Schiavone¹, Stéphanie P.Lacour¹

¹ Laboratory for Soft Bioelectronic Interfaces, EPFL, Lausanne, Switzerland
² IT'IS Foundation, Zurich, Switzerland
³ Integrated Systems Laboratory, ETH Zurich, Zurich, Switzerland

1 Motivation and goal

Hybrid integration referred to the co-design, integration, and synergistic function of thin-film elements, thicker films, and active electronic components - in a cohesive system. We aim at realizing **a completely wireless interface** leveraging microfabrication and thin-film technology to realize the RF elements for wireless power transfer and communication.

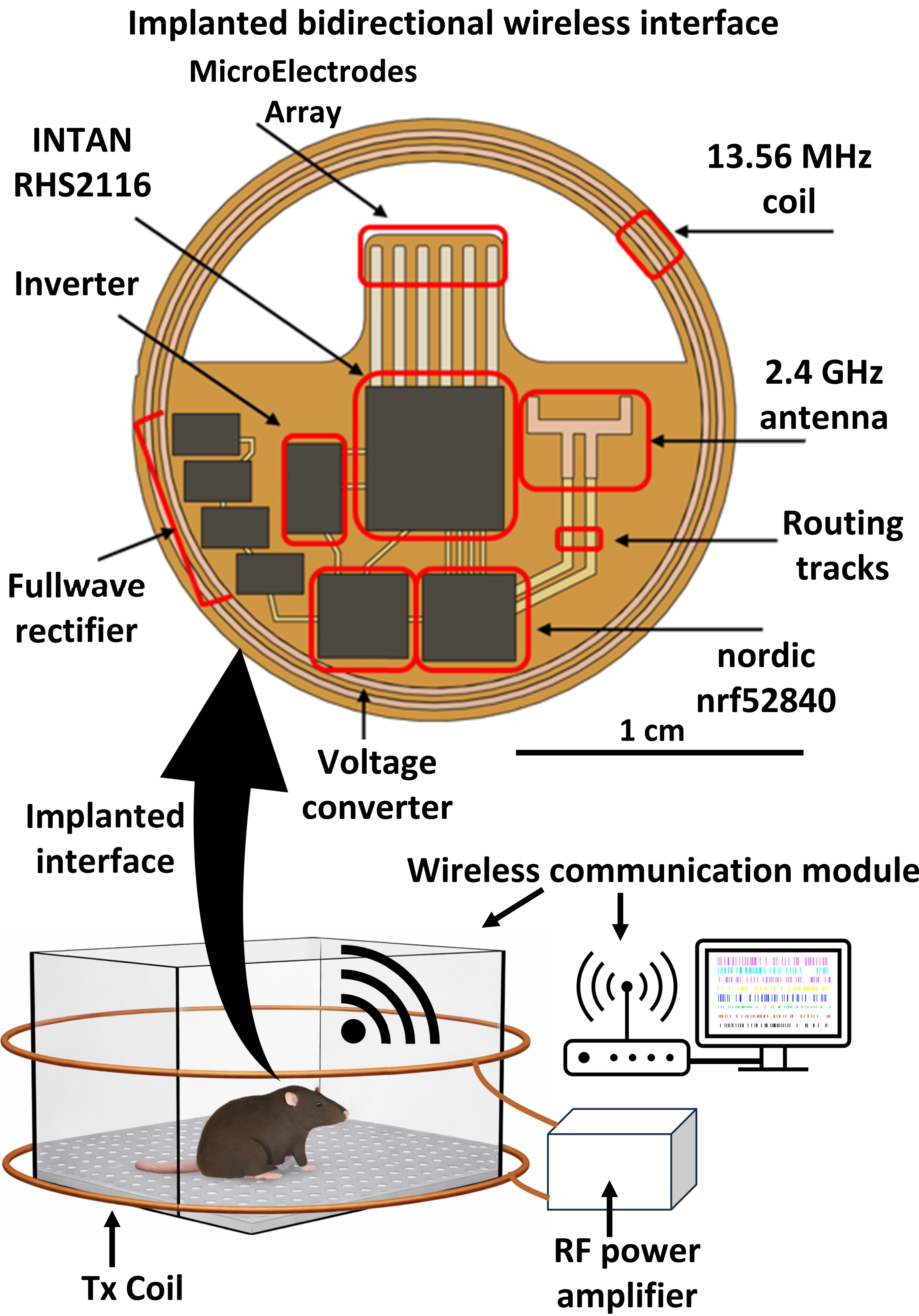
Current state of the art

- Limited number of channels:** limited by required feedthroughs
- Increased system failure points:** different mechanical properties between blocks
- Increased system dimensions:** form factor is a key parameter for the inflammation response

Challenges

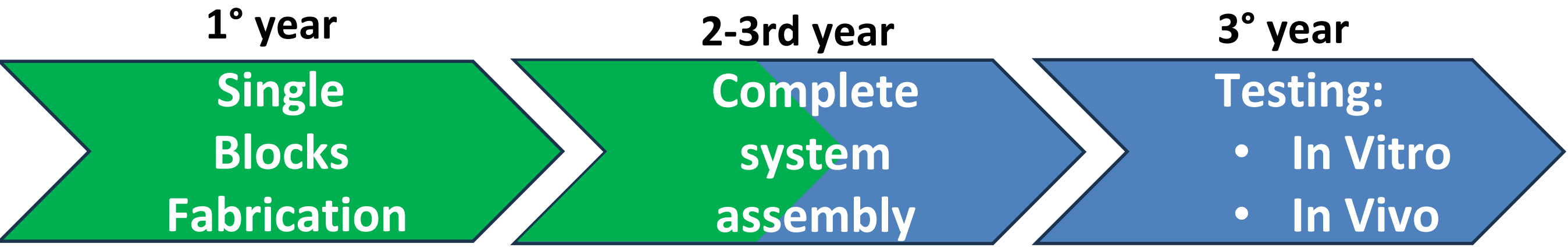
- Integration** of bare dies electronics onto thin films substrates (chip-to-flex bonding)
- Realization** of thin-film RF elements, wireless power transfer and communication

2 Proposed solution



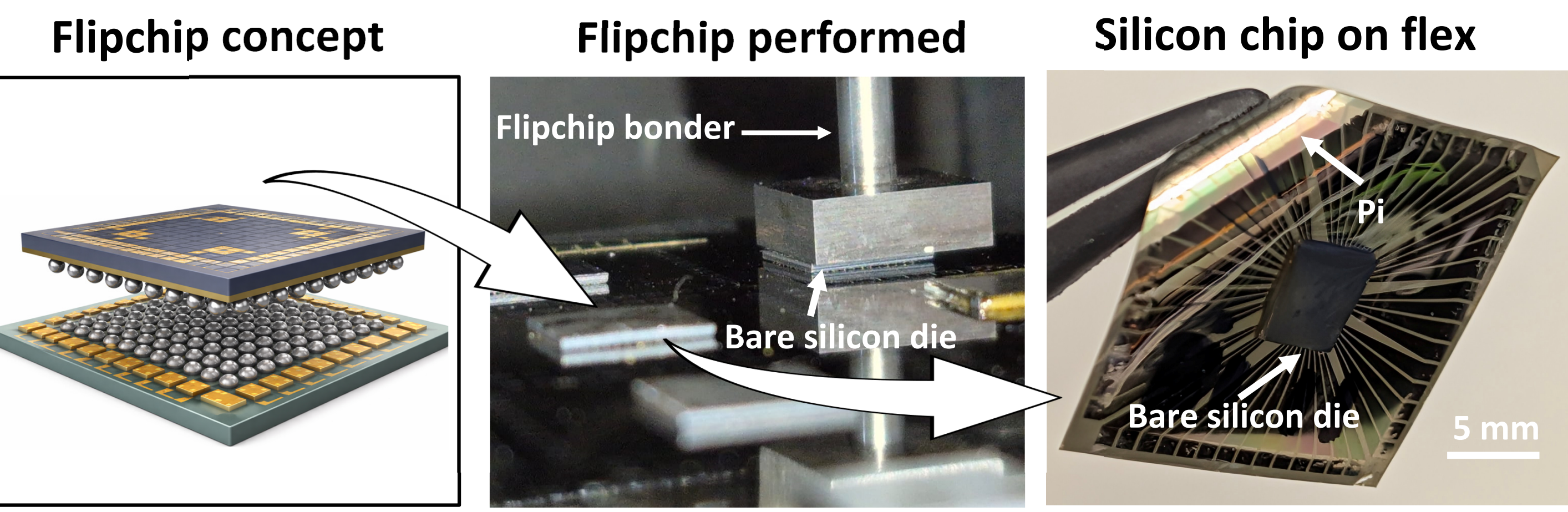
Proposed implanted bidirectional wireless interface. The system integrates a 13.56 MHz inductive power link, on-chip power management, a 2.4 GHz Bluetooth Low Energy antenna, and a neural interface based on an INTAN chip, enabling fully wireless in-vivo operation.

5 Outlook and conclusions



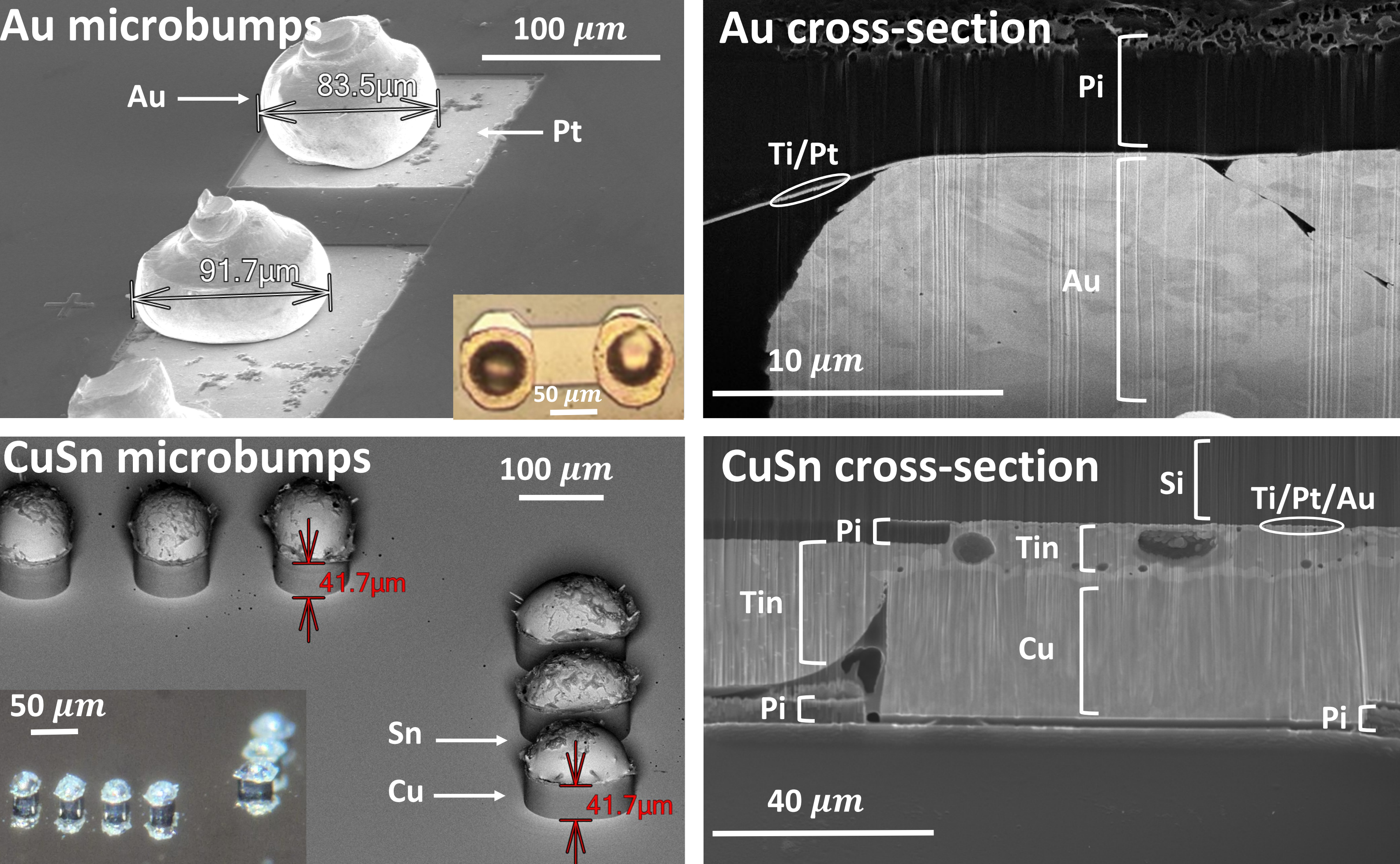
This work aims at demonstrating a proof of concept for a fully wireless neural interface using hybrid integration technology, achieved through single-wafer integration and to be validated through in-vivo functional testing.

3 Technology driver 1: chip-thin substrate bonding



Flip-chip integration on thin substrate (8 μm): schematic concept (left), bonding process (center), and bonded silicon die on polyimide substrate (right)

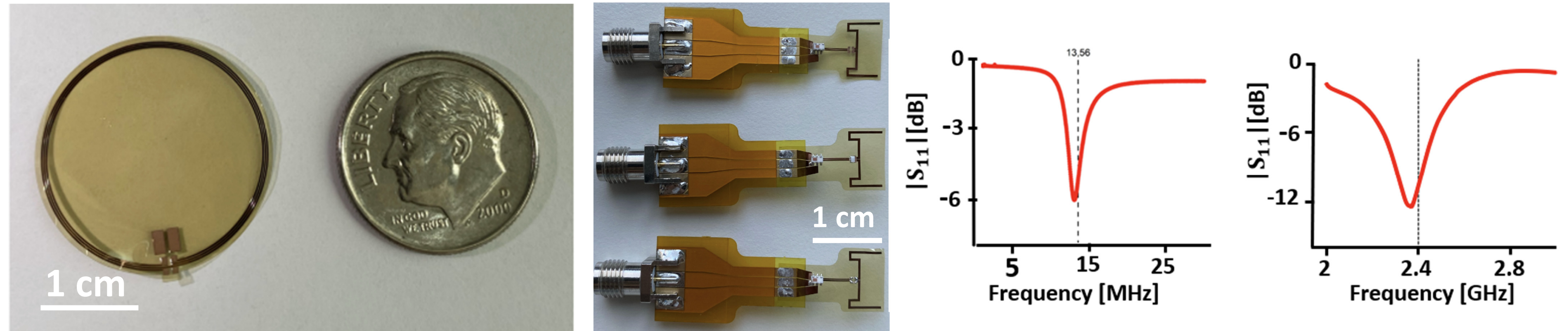
Comparison between Au and CuSn bumps



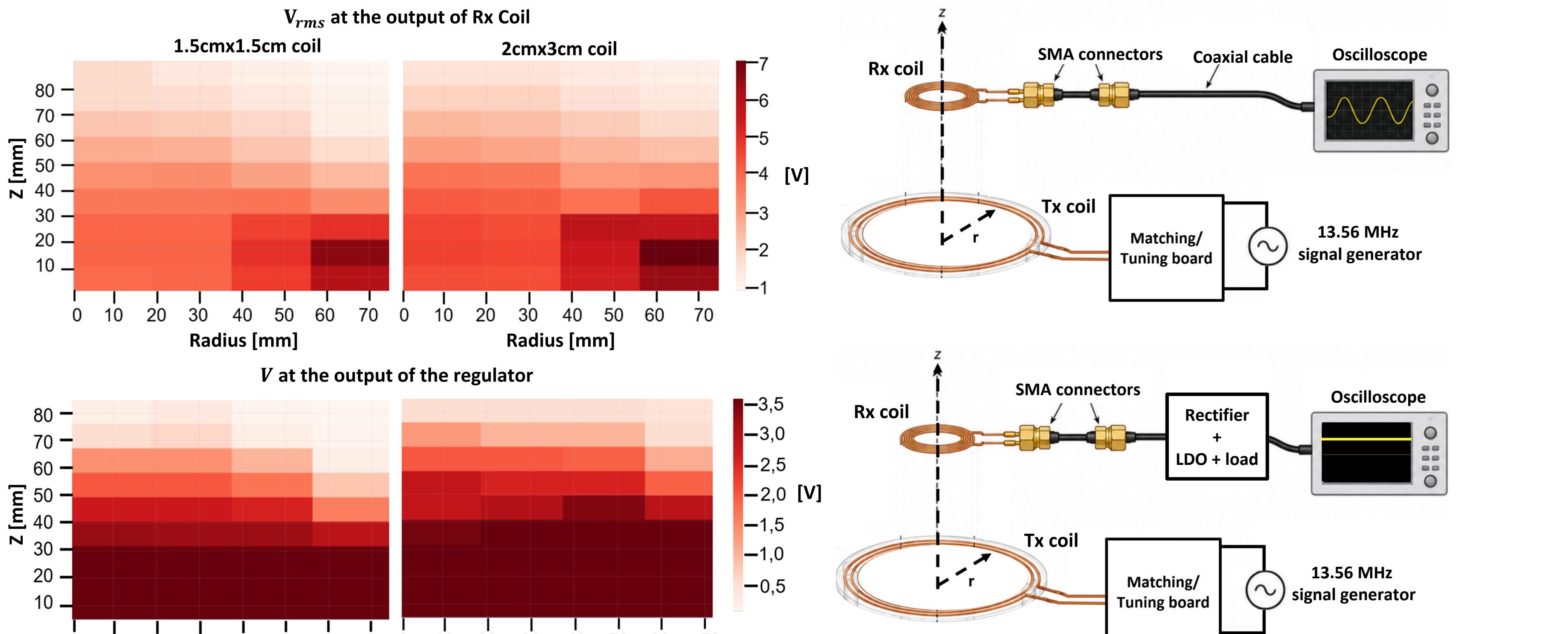
Material	Ω/bump	Yield(%)	Pros	Cons
Au	5 – 10	90 – 95	Simple	Non – Scalable
CuSn	0,5 – 1	98 – 99	Scalable	Complex Fab

CuSn microbumps provide higher scalability and yield than Au microbumps, which are limited to low pin counts (<70) due to excessive bonding forces on thin films.

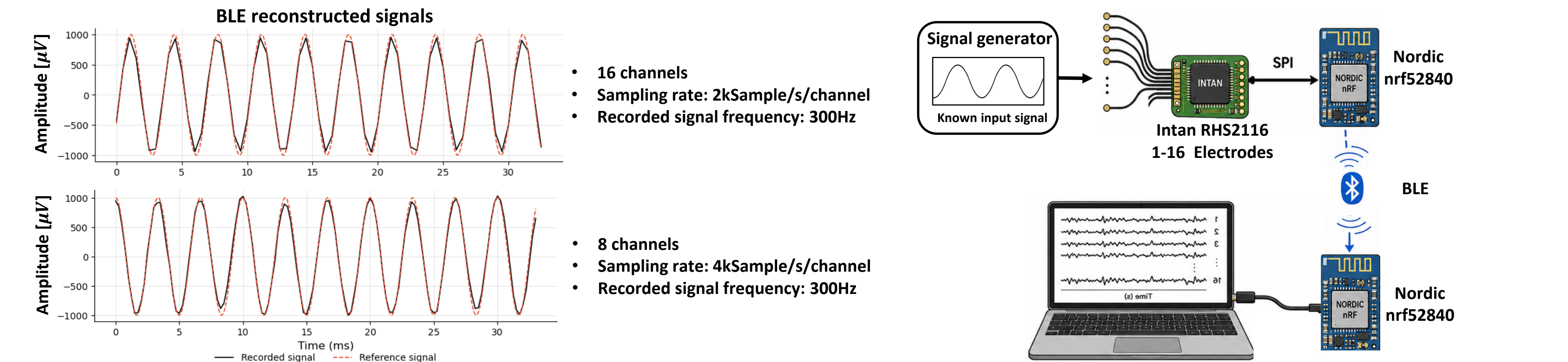
4 Technology driver 2: RF elements, WPT and BLE



Fabricated RF elements. Electroplated Cu inductive coil for 13.56 MHz wireless power transfer and antennas for 2.45 GHz wireless communication. S11 measurements confirm the resonance of the coil at 13.56 MHz and the antenna response around 2.4 GHz. Structures were fabricated by 8 μm Cu electrodeposition on polyimide substrates.



WPT cage characterization. Spatial mapping of the received voltage for two Rx coil geometries inside the Tx coil volume. Measurements were performed first at the open-circuit Rx output and then after rectification and LDO regulation, to assess the availability of a stable 3.6 V supply across different radial and vertical positions.



BLE communication. A known 300 Hz sinewave was injected into the INTAN inputs, digitized by the INTAN front-end, transferred to the Nordic nRF52840 through SPI, and transmitted via BLE to a second Nordic receiver connected to the PC. The reconstructed signals match the reference waveform, demonstrating wireless recording from 1 up to 16 channels.